

Precision and Recall

Project: Responsible Plant Classifier

1. Precision and Recall

Teach the learning cycle

Primary target: Choose the denominator that answers the question

Session · Sequence and minutes · Resources

Session: Session 1 · Understand · 45 min

Sequence and minutes: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Sequence and minutes: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Sequence and minutes: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building. Repeated copies of a case share one recorded attempt; they are not new evidence.

Retrieval answer guidance

Accuracy and Imbalance

Without opening the old lesson, explain one key idea from Accuracy and Imbalance. Give an example and a mistake that the idea helps you avoid.

Look for: Explain why overall accuracy can hide failure on a less common class

Reference example, not the only acceptable wording: The training-majority rule always predicts clear. Matrix $\begin{bmatrix} 16 & 0 \\ 4 & 0 \end{bmatrix}$ gives $16/20=80$ percent, $FP=0, FN=4$. It misses all four blocked cases, so the overall score hides complete failure on that evaluated class.

Scores and Calibration Intuition

Without opening the old lesson, explain one key idea from Scores and Calibration Intuition. Give an example and a mistake that the idea helps you avoid.

Look for: Interpret calibration using groups of forecasts

Reference example, not the only acceptable wording: Teacher explanation: $2/10=.2$ and $6/10=.6$.

Decision Trees

Without opening the old lesson, explain one key idea from Decision Trees. Give an example and a mistake that the idea helps you avoid.

Look for: Fit a root question and leaf predictions from training examples

Reference example, not the only acceptable wording: The root asks near, with branches 0 and 1. Leaves predict clear for 0 and blocked for 1. Depth is one. Training labels determined both leaf majorities and the error comparison that selected near rather than tall.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

$TP=9, FP=3, FN=6, TN=12$. Compute precision and recall. What happens to precision when a system predicts no positives?

Concept to explain

Precision measures how many positive predictions are right; recall measures how many actual positives are found.

Term · Meaning

Term: precision

Meaning: TP divided by TP plus FP

Term: recall

Meaning: TP divided by TP plus FN

Term: false alarm

Meaning: false-positive prediction

Term: miss

Meaning: false-negative prediction

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Teach two 45-minute sessions. This week focuses on precision recall and denominators. Model the worked example first, collect the new W 4 case independently before feedback, and use the fresh retry to check transfer. Award credit for traceable reasoning and bounded claims, including justified uncertainty.

Readiness answer

The common class can dominate the total; inspect the full confusion matrix and class counts.

Printed student lesson and workbook

Paper, pencils and a calculator for fractions

The fictional examples and a blank evidence table

Optional Python 3 editor; no external packages or AI accounts required

End goal for the pictorial case

By the end, I can explain precision and recall using a traced example and a stated limitation.

This added worked case illustrates the week's main idea. Retain all original and connected topics;

schedule its practice within the teaching cycle.

2. Explain the mechanism

Explain the mechanism

Trace the information, fitted behavior, and evaluation proof separately.

QUESTION Calculate both and choose a priority for two consequence scenarios.

MODEL IDEA Precision measures how many positive predictions are right; recall measures how many actual positives are found.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD Confusion counts: TP=8, FP=2, FN=4, TN=16. Precision = $8 \div (8+2) = 80\%$.

Draw a fresh data → model → prediction → evaluation trace:

Predict first. Then check each step. A correct answer without visible evidence cannot be audited.

STEP 1 Confusion counts: TP=8, FP=2, FN=4, TN=16.

STEP 2 Precision = $8 \div (8+2) = 80\%$.

STEP 3 Recall = $8 \div (8+4) \approx 67\%$.

STEP 4 Choose threshold and human review by comparing the cost of two false alarms with four misses.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the evidence independently:

Concrete teaching case

Positive is spotted: TP=8, FP=2, FN=4.

Compute precision and recall.

Exact explanation to model

Precision=8/10=80%; recall=8/12 about 66.7%. Denominators are predicted positives and actual positives respectively.

Explain the added worked example

Blocked remains positive. Precision asks: of the cases flagged blocked, how many were actually blocked according to the reference? Recall asks: of all reference-blocked cases, how many were found? Both use true positives, but they answer different questions. Keep reference rows and prediction columns ordered clear then blocked.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data

split visible as you reason. Earlier weeks supplied the model-building process; this week examines precision recall and denominators.

For matrix $\begin{bmatrix} 12 & 4 \\ 1 & 3 \end{bmatrix}$, $TN=12$, $FP=4$, $FN=1$, $TP=3$. There are seven predicted positives and four reference positives. Precision is $3/7$, about 42.9 percent. Recall is $3/4=75$ percent. Accuracy is $15/20=75$ percent. A correct denominator matters more than memorizing a percentage.

Trace the calculation on paper before running this optional Python check. Use the stated fictional inputs and compare intermediate values. The code checks this example; your independent workbook case uses different evidence.

Key terms

Term · Meaning

Term: Precision

Meaning: True positives divided by all predicted positives

Term: Recall

Meaning: True positives divided by all reference positives

Term: Denominator

Meaning: The group counted in the bottom of a fraction

Term: Undefined

Meaning: A ratio with no valid value because its denominator is zero

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a prediction, then explain the deciding step. These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Precision recall and denominators at a glance

Part · Evidence or action · Interpretation

Part: Precision

Evidence or action: 3 correct flags

Interpretation: 7 total flags

Part: Recall

Evidence or action: 3 found blocked cases

Interpretation: 4 reference blocked

Part: Accuracy

Evidence or action: 15 correct labels

Interpretation: 20 total records

Draw a box around the predicted-blocked column for precision and around the reference-blocked row for recall. Their overlap is TP. A model can flag nearly everything and find most blocked cases while producing many false alarms. Conversely, a model can flag very few cases accurately while missing many positives.

Calculate precision, recall and accuracy for the worked matrix. Explain the steps or evidence behind your response.

Thinking move · Teacher prompt

Thinking move: Plan

Teacher prompt: What is given? What is the question? Which procedure fits these facts?

Thinking move: Do

Teacher prompt: Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.

Thinking move: Check

Teacher prompt: Does the output answer the question? Which assumption or limitation prevents a larger

claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Precision and Recall

Put precision and recall denominators on the picture instead of memorising names.

The facts we will use

Positive means spotted. TP=6, FP=2, FN=4, TN=8 in twenty fictional leaves.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/17

Keep precision and recall denominators distinct

	predicted spotted	predicted healthy
actual spotted	6	4
actual healthy	2	8

Read a cell using both its row and column labels.

Precision and Recall

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Predicted positives

Eight leaves were flagged: $\text{precision} = 6/(6+2) = 75\%$.



2. Actual positives

Ten leaves were spotted: $\text{recall} = 6/(6+4) = 60\%$.



3. Keep goals separate

Precision asks how reliable flags are; recall asks how many spotted leaves were caught.

CHANGE ONE THING

Correct two false negatives into true positives.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Predicted positives

What happens and why: Eight leaves were flagged: $\text{precision} = 6/(6+2) = 75\%$.

Step: 2. Actual positives

What happens and why: Ten leaves were spotted: $\text{recall} = 6/(6+4) = 60\%$.

Step: 3. Keep goals separate

What happens and why: Precision asks how reliable flags are; recall asks how many spotted leaves were caught.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Read the picture in order. Precision asks how reliable flags are; recall asks how many spotted leaves were caught. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

If a relevant denominator is zero, report that metric as undefined under the chosen convention rather than silently inventing success.

Change one thing in the pictured case

Change: Correct two false negatives into true positives.

Prediction: Precision becomes $8/10=80\%$; recall becomes $8/10=80\%$.

Reason: TP becomes 8 and FN 2; FP remains 2, so the two denominators happen to equal 10.

Ask while showing the picture

Cover the result. Ask learners to name the inputs and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its evidence. Then change the supplied condition and compare. The picture is a representation; it does not document execution of a real AI model.

3. Demonstrate and guide practice

Demonstrate and guide practice

Board sequence

Write the given facts: Positive is spotted: TP=8, FP=2, FN=4.

Write the question to explain: Compute precision and recall.

Reveal after predictions: Precision= $8/10=80\%$; recall= $8/12$ about 66.7%. Denominators are predicted positives and actual positives respectively.

Keep for an independent attempt: With disease positive, TP=6, FP=2 and FN=4. Calculate precision and recall, showing both denominators.

Classroom dialogue

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks evidence; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Guided response: Precision $3/7$ about 42.9 percent, recall $3/4=75$ percent, accuracy $15/20=75$ percent.

Exit response: Request TP, FP, FN, sample size, positive class and precision. Finding every positive can coexist with many false alarms.

Case-specific teaching sequence

Stage · Teacher action

Stage: Session 1 start to 7 minutes

Teacher action: Use the prerequisite question: Why can 80 percent accuracy hide missed blocked cases? Check the explanation before proceeding.

Stage: 7 to 18 minutes

Teacher action: Read the opening and model the worked evidence. Ask students to identify precision in context.

Stage: 18 to 32 minutes

Teacher action: Complete W 1 and W 2 with support: identify the groups; calculate both ratios. Require the evidence, not only the final response.

Stage: 32 to 45 minutes

Teacher action: Read the visual row by row. Collect the guided explanation and compare reasoning in pairs. Trace the Python check after the paper calculation.

Stage: Session 2 start to 8 minutes

Teacher action: Retrieve the distinction between precision and recall. Invite a counterexample to an overconfident claim.

Stage: 8 to 20 minutes

Teacher action: Discuss the application and complete W 3: Explain precision and recall for $[[16, 0], [4, 0]]$.

Stage: 20 to 35 minutes

Teacher action: Collect W4 independently before giving hints. Assess these outcomes: Choose the denominator that answers the question; Calculate precision and recall with a declared positive class. Record any support used.

Stage: 35 to 45 minutes

Teacher action: Use W 5 and the exit response to assess limits. Give targeted feedback, then the fresh retry. Save evidence of independent understanding.

Reduce help deliberately

Completion task: Calculate precision, recall and accuracy for the worked matrix.

Teacher check: Teacher explanation: Precision $3/7$ about 42.9 percent, recall $3/4=75$ percent, accuracy $15/20=75$ percent.

Start by identifying the given inputs together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood. For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Precision recall and denominators. Proposed explanation: "A final answer is enough". Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Ask for the relevant counts, units, procedure or saved record. For this week: Predicted positives $TP+FP=7$; reference positives $TP+FN=4$.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class. If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Were the original precision and recall denominators the same? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

Reasoned teacher answer

No. Precision used 8 predicted-positive leaves; recall used 10 actual-positive leaves. Their meanings remain distinct even when numbers later coincide. Changed-case check: Precision becomes $8/10=80\%$; recall becomes $8/10=80\%$. TP becomes 8 and FN 2; FP remains 2, so the two denominators happen to equal 10.

Changed-case reasoning: Precision becomes $8/10=80\%$; recall becomes $8/10=80\%$. TP becomes 8 and FN 2; FP remains 2, so the two denominators happen to equal 10.

Collect each learner's explanation before discussion. Use the first missing connection between input, deciding step and result to choose support. This question is worked-case practice, not fresh mastery evidence.

4. Run the weekly evidence lab

Run the weekly evidence lab

Keep the test conditions fixed, record every result, and change only one planned factor.

1. Write the hypothesis and expected evidence before running.
2. Metric Tradeoff Lab: compute both metrics for two thresholds, map errors to consequences, and defend a choice.
3. Run one normal, one boundary, one missing, and one shifted case.
4. Calculate the metric from raw counts or row-level errors.
5. Interpret what changed, what did not, and what remains unsafe or unknown.

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing. Keep the expected result separate from an observed mistake.

Record · What to capture

Record: Input or evidence

What to capture: Exact data, instruction or claim

Record: Expected result

What to capture: Answer or safe fallback declared before the run

Record: Observed result

What to capture: Actual result, including errors

Record: Explanation

What to capture: Rule, calculation or evidence explaining the outcome

Record: Next test

What to capture: One justified change and a new prediction

Diagnose and reteach

MISCONCEPTION TO REPAIR Precision and recall are two names for the same accuracy calculation.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation. Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

A final answer is enough: Ask for the relevant counts, units, procedure or saved record. For this week: Predicted positives $TP+FP=7$; reference positives $TP+FN=4$.

A stronger claim makes a stronger project: Request TP, FP, FN, sample size, positive class and precision. Finding every positive can coexist with many false alarms.

The new case can be copied from the example: The independent task changes the evidence. Require a fresh explanation: $TP=6$, $FP=2$, $FN=3$. Precision $6/8=75$ percent; recall $6/9$ about 66.7 percent; accuracy $14/19$ about 73.7 percent.

Match the support to the first error

What the response shows · Next teaching action

What the response shows: Missing or misread facts

Next teaching action: Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

What the response shows: Wrong procedure

Next teaching action: Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.

What the response shows: Arithmetic or implementation error

Next teaching action: Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows: Correct result without a reason

Next teaching action: Ask for a trace and a changed case before marking independent understanding.

What the response shows: An unsupported wider claim

Next teaching action: Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal. Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

5. Assess and explain the answers

Assess and explain the answers

Worked case answer

Precision= $8/10=80\%$; recall= $8/12$ about 66.7%. Denominators are predicted positives and actual positives respectively.

Independent question

With disease positive, TP=6, FP=2 and FN=4. Calculate precision and recall, showing both denominators.

Expected independent answer

Precision is $6/(6+2) = 75\%$. Recall is $6/(6+4) = 60\%$. Precision counts predicted positives; recall counts actual positives.

What counts as evidence

Criterion · 0 points · 1 point · 2 points

Criterion: Result

0 points: Incorrect or absent

1 point: Partly correct

2 points: Correct with necessary conditions

Criterion: Reasoning

0 points: No supporting trace

1 point: Some steps

2 points: Clear mechanism, calculation or source support

Criterion: Limit

0 points: Overclaims certainty

1 point: Vague limitation

2 points: Relevant uncertainty or missing fact

Criterion: Transfer

0 points: Copies without adapting

1 point: Adapts with prompts

2 points: Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold. Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the evidence that challenges this claim: “Precision and recall use the same denominator.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Precision= $9/12=75\%$; recall= $9/15=60\%$. With no predicted positives, $TP+FP=0$ so precision is undefined mathematically; reporting software may apply a convention that must be disclosed. Do not interpret a zero-

denominator convention as observed performance.

Explanations for the additional workbook examples

W1 Identify the groups

For the worked matrix, give predicted-positive and reference-positive totals.

Teacher explanation: Predicted positives $TP+FP=7$; reference positives $TP+FN=4$.

Predicted positives include true positives and false positives, so add $TP + FP$ to get 7. Reference positives include true positives and false negatives, so add $TP + FN$ to get 4. These are different groups: one is defined by the prediction, the other by the reference label.

W2 Calculate both ratios

Calculate precision, recall and accuracy for the worked matrix.

Teacher explanation: Precision $3/7$ about 42.9 percent, recall $3/4=75$ percent, accuracy $15/20=75$ percent.

W3 Handle no flags

Explain precision and recall for $[[16, 0], [4, 0]]$.

Teacher explanation: Precision is undefined, with no flags; recall is $0/4=0$. This does not mean excellent detection.

W4 Apply independently

Use $[[8, 2], [3, 6]]$. Give TP, FP, FN, precision, recall and accuracy.

Teacher explanation: $TP=6$, $FP=2$, $FN=3$. Precision $6/8=75$ percent; recall $6/9$ about 66.7 percent; accuracy $14/19$ about 73.7 percent.

W5 Limit a claim

A team reports recall 100 percent but hides precision and counts. What is missing?

Teacher explanation: Request TP, FP, FN, sample size, positive class and precision. Finding every positive can coexist with many false alarms.

Assess independent transfer

Selected case: Use $[[8, 2], [3, 6]]$. Give TP, FP, FN, precision, recall and accuracy.

Selected case answer

Teacher explanation: $TP=6$, $FP=2$, $FN=3$. Precision $6/8=75$ percent; recall $6/9$ about 66.7 percent; accuracy $14/19$ about 73.7 percent.

Dimension · Evidence required

Dimension: Explanation

Evidence required: Names the applicable idea or procedure and ties it to the supplied facts.

Dimension: Trace and result

Evidence required: Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.

Dimension: Limits

Evidence required: States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace. Do not penalise an honestly recorded model failure if the learner correctly explains its evidence and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case. Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

6. Connect and extend learning

Connect and extend learning

Responsible Plant Classifier

Set a Plant Classifier review threshold and explain which error type it prioritizes and who makes the final decision.

Keep the decision, input, result, explanation and remaining limitation in the project evidence record. Retrieve this record next week rather than restarting the project.

Support and stretch

Use a tiny labelled dataset, explicit units and a named positive class. Stretch: require a baseline, held-out evaluation and subgroup or shifted-condition analysis with denominators.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more evidence or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 18: Error Analysis. Retrieve today's evidence and identify what carries forward and what is new.

Support and fresh reassessment

Provide a blank evidence table and allow a verbal explanation before writing. Highlight the given conditions without supplying the W 4 conclusion. Use precision and recall in full sentences. Extension: alter one condition in the worked example and explain which conclusion changes and which remains supported.

For independent W 4, require: TP=6, FP=2, FN=3. Precision $6/8=75$ percent; recall $6/9$ about 66.7 percent; accuracy $14/19$ about 73.7 percent. Fresh retry $[[5, 1],[2, 4]]$: precision $4/5=80$ percent, recall $4/6$ about 66.7 percent, accuracy $9/12=75$ percent. Require both denominators. Mark each criterion as independent, supported or not yet demonstrated. Reassess the reasoning after feedback; do not replace the first attempt silently.

Next week: Thresholds and consequences. Carry forward the saved evidence and the distinction between development choices and justified claims.

Plan the delayed return

Suggested return points in this level: ai-4/18, ai-4/20, ai-4/23. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used. For a new level, revisit the previous capstone before checking the new prerequisites.

Record · Teacher evidence

Record: Learner code and date

Teacher evidence: _____

Record: Concept and case used; prior exposure

Teacher evidence: _____

Record: Explanation / trace / limits

Teacher evidence: _____

Record: Support and first repair

Teacher evidence: _____

Record: Changed case and delayed outcome

Teacher evidence: _____

Record: Next teaching action

Teacher evidence: _____

Use the evidence to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.